Mathematical overview

Independent uniform replacement draws from A=(2,4,9), B=(1,6,8), C=(3,5,7) give 27 equally likely simultaneous triples. A and B each supply the largest card in 10 triples; C in 7. A pairwise majority cycle does not imply equal chances in a three-way contest.

Two replacement draws per player give nine equally likely ordered pairs per deck and 81 outcomes per comparison. A/B has wins 37/44 with no ties; B/C and C/A each have wins 39/38 with four ties. Scoring two for a win and one for a tie reverses A's original advantage over B. Equal sums have multiplicities; different sum values are not equally likely.

Concealed deck choice: against fixed A,B,C, pure A averages (1,10/9,8/9) points, B (8/9,1,10/9), C (10/9,8/9,1). Equal mixing earns exactly one against every independent opponent mixture. Every policy's three fixed-opponent scores average one, so no policy exceeds one against all three. Equal label proportions are the unique policy attaining the best worst-case average. These are exact all-outcome averages, never a promise about a finite play sample.

Children compare a simultaneous champion, a changed contest, and a concealed policy: three investigations. Grades 2-3 can construct and sort physical outcomes; Grades 4-5 can certify counts and policy guarantees. An optional supported K-1 champion comparison needs only number ordering. Trials test ideas; complete counts and equalizing arguments establish the facts. Uniform slips/cards, replacement and private independent draws are essential.

Entry routes and readiness

Fixed tables: K–1 / 2–3 / 4–5, with one anchored adult each. Use one investigation and its brief launch per table visit. Adults read/record while children own the mathematical choices. P1 needs comparisons to 9; P2-3 addition through 18 and outcome multiplicity; P4-5 private independent choice and exact versus sample averages. Supported K-1 champion work is optional; fractions are adult shorthand.

Materials and preparation

Per pair/trio: six labeled opaque bags, two independent copies each of A,B,C (18 identical-backed 4 by 6 cm cards); two private label bags; six initial same-size slips and twelve blanks; counters, pencils and paper. Same-label play needs separate copies. Reference distributions stay visible; cards/slips must not be identifiable by touch.

Per table, prepare nine reusable two-draw outcome cards per deck: first value, second value, total. Retain physical multiplicities when grouping totals. Pairs divide comparisons and pool a checkable record, avoiding repeated transcription. Preparation: about 10-15 minutes. Mixing, independence and handling are unrehearsed.

Launch together

Allow brief card handling. Lay out a practice triple and name its largest card; show returning cards and mixing separately. For the two-draw visit, show replacement between draws before using the outcome cards. For concealed choice, replay the printed label B -> deck B -> card 6 example against A -> 4: two points and zero. This demonstrates the action, not a policy's average.

First visit and return visits

Allow 2-3 minutes to launch and 20-35 minutes for P1 or P2-3. Return with saved nine-card distributions for P3, or spend 20-40 minutes on P4 before P5. These timings are proposals. Pool complete casework at the table; retain physical records and give pages by readiness.

Problem 1: Three-way champion

Entry, prerequisites and timing

Compare one visible triple, then let pairs choose how to assemble complete cases. Grades 2-3 can share the 27 cases across a table; Grades 4-5 can count by the potential winning card. Allow 15-25 minutes or continue on a later visit.

Mathematical facts and solutions

There are 3 times 3 times 3 = 27 triples. No two decks share a value, so there are no ties. For a possible winning card x, multiply the number of smaller cards in the two other decks. A's cards 2,4,9 win 0,1,9 triples; B's 1,6,8 win 0,4,6; C's 3,5,7 win 1,2,4.

Therefore A wins 10, B wins 10, C wins 7. A and B tie for greatest chance, 10/27 each. The three counts sum to 27, a completeness check. This simultaneous winner is a different question from the base's three pairwise advantages. A short run of games need not have these proportions.

Hold-back hints

After an attempt, hold one deck's card fixed and ask which choices of the other two cards leave it largest. Use nine movable pairs or a shared list rather than asking each child to copy every case.

Depth and further questions

Why does A's 9 always win, while A's 4 wins only once? Explain each product with actual pairs. Let children propose a different three-way scoring rule, but solve its exact model before claiming any new probabilities.

Facilitator notes

Preserve triples and their winners in one pooled record. Adult recording support should leave case selection and checking with the children.

Problem 2-3: Totals can change an advantage

Entry, prerequisites and timing

P2 starts with actual replacement draws and sums, then sorting the nine outcome cards per deck. P3 requires keeping frequencies or distinct physical cards visible. Allow 20-40 minutes; the exact comparisons may be a whole return visit.

Mathematical facts and solutions

P2, first-card row order as printed: A totals are (4,6,11), (6,8,13), (11,13,18); B totals (2,7,9), (7,12,14), (9,14,16); C totals (6,8,10), (8,10,12), (10,12,14). Each row contains the three possible second draws. Reversing distinct draws creates a second equally likely outcome even when its total is unchanged.

Total frequencies: A 4:1, 6:2, 8:1, 11:2, 13:2, 18:1; B 2:1, 7:2, 9:2, 12:1, 14:2, 16:1; C 6:1, 8:2, 10:3, 12:2, 14:1. Each frequency list totals nine. Compare nine physical outcomes with nine, or multiply grouped frequencies for each total comparison.

P3: A/B gives A 37 wins, B 44, no ties. Total points over 81 outcomes are 74 and 88, so averages 74/81 and 88/81: B is better. B/C gives 39 B wins, 38 C wins, four ties; total points 82 and 80, averages 82/81 and 80/81: B is better. C/A gives 39 C wins, 38 A wins, four ties and the same point totals: C is better. Every pairing totals 162 points over 81 rounds.

The one-draw A-over-B advantage reverses; the other two directed advantages remain under this tie score. Summing two draws changes the distribution and the contest. Equal deck totals or an equal average number do not settle which distribution wins more comparisons.

Hold-back hints

Offer grouped total piles after children have explored their own organization. Compare one A total with all B totals, counting physical multiplicities. Keep the nine reusable cards intact so an equal-total pile does not become one equiprobable outcome.

Depth and further questions

Explain why duplicating all nine outcome cards equally preserves the exact average, but retaining only one card per different total can change it. Check the frequency sums and the two-points-per-round total as independent audits of a proposed comparison.

Facilitator notes

Pool the three comparisons across pairs. A reliable complete collection is mathematical work; copying it independently three times is not required. Do not silently discard ties.

Problem 4-5: Hide the choice; equalize its risks

Entry, prerequisites and timing

Play legal independent rounds before calculating. Every label bag is nonempty; an individual label may have zero slips. Grades 2-3 can test bags and compare pooled exact scores; Grades 4-5 can explain why no bag beats one against every fixed opponent. Allow 20-40 minutes and return for the proof.

Mathematical facts and solutions

P4: pure A scores (1,10/9,8/9) against A,B,C; pure B (8/9,1,10/9); pure C (10/9,8/9,1). The same-deck average is one: each nine-pair self-comparison has three wins, three losses and three ties. An A,B,C label bag therefore averages one against every fixed deck. Any equal positive number of each label gives the same policy.

For any child-designed bag with proportions a,b,c summing to one, its exact averages against A,B,C are 1+(c-b)/9, 1+(a-c)/9, 1+(b-a)/9. Multiply a pure row by its label frequency and add. For example, a pure-A bag averages 8/9 against C; adding hidden variety is not automatically an improvement unless the worst score improves.

P5: no bag earns more than one against all three. The three exact averages sum to three, so at least one is at most one. If all are at least one, then c>=b, a>=c, b>=a, forcing a=b=c. Equal mixing uniquely maximizes the worst-case exact average. Against any independent opponent mixture it still earns one, by averaging its three fixed-deck scores.

This guarantee requires the opponent not seeing the realized private label before choosing its own label. It is an expectation over all label-slip and card draws, not a claim that one finite play sample will average exactly one. Replacement restores the same policy each round.

Hold-back hints

Challenge a proposed bag with each fixed opponent separately. Keep three score columns. For the upper bound, ask what their three averages add to rather than directing a search through many larger bags.

Depth and further questions

Compare a bag's best opponent-specific score with its worst score. If the realized label is revealed before the opponent chooses, the opponent can always select its conqueror, lowering the chooser's expected score to 8/9: the information rule has changed.

Facilitator notes

Use independent copies for same-deck draws. A shared label slip or coordinated shuffling changes the probability model. A short game is an experiment; exact outcome weighting explains the policy.

Sources, checks and classroom record

Sources and adaptation

Current Week 24 student packets and adult guide establish the fair-card model and the 5/9 pairwise cycle. The base outline cites Conrey et al., Intransitive Dice (2016), pp. 1-2, for model background; no new reading of that paper or attribution of these bonus conclusions is claimed.

Rozhkovskaya, Math Circles for Elementary School Students, Lesson 13, Gnomes and At the lesson (downloaded EPUB OEBPS/part0023.xhtml), was read: it reports children coordinating or altering desired outcomes and doubts about fairness. Our explicit independence, replacement and count-versus-trial distinction are proposed adaptations. Related AP-22 has different unequal two-choice payoffs.

Verification

The fresh student math review independently enumerated all 27 triples and 81 outcomes per summed comparison, checked the pure payoff matrix and exact mixture argument, and located the now-corrected empty-label-bag edge case. These guide answers use the revised final student packet and synchronized answer notes.

The final guide received a separate independent mathematical review. Both final PDFs were rendered and inspected and passed a clean extracted-source ZIP rebuild. See release-validation.json in the editable source package. Digital checks do not establish physical bag fairness, mixing, independent choice or classroom pacing; no physical pretest or classroom pilot has occurred.

What to record after a visit

Record which of the three investigations each pair actually used, saved distributions or bags, the child's conjecture and a useful next question.

Observe whether reverse draws retain multiplicity, exact averages are confused with observed scores, private choices remain independent, and children organize the cases themselves. Separate these observations from hypotheses about reading or readiness.

Status

Draft bonus companion. Unpiloted. Digital and mathematical checks do not establish physical fit, procedural rehearsal, staffing success or classroom pacing. Physical pretests have not been performed. Base student pages and base facilitator guide are separate.